INCH-POUND

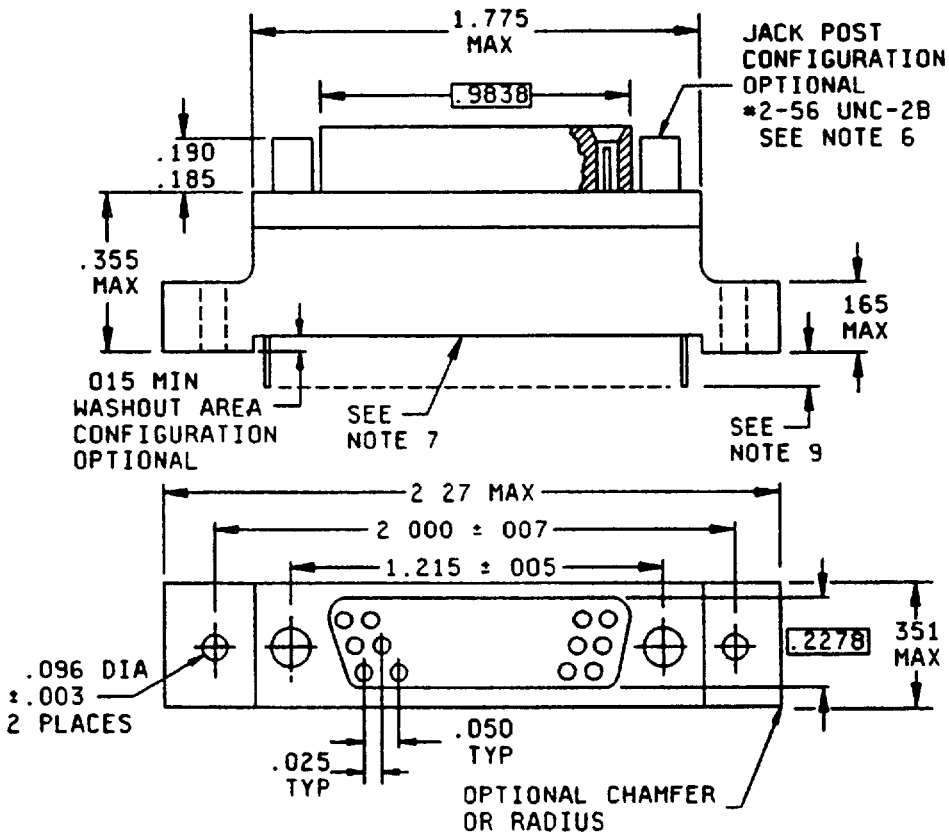
MIL-C-83513/23A
14 Oct 1993
SUPERSEDING
MIL-C-83513/23
11 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, STRAIGHT, PIN CONTACTS, 3 ROW, SOLDER TYPE, 51 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.

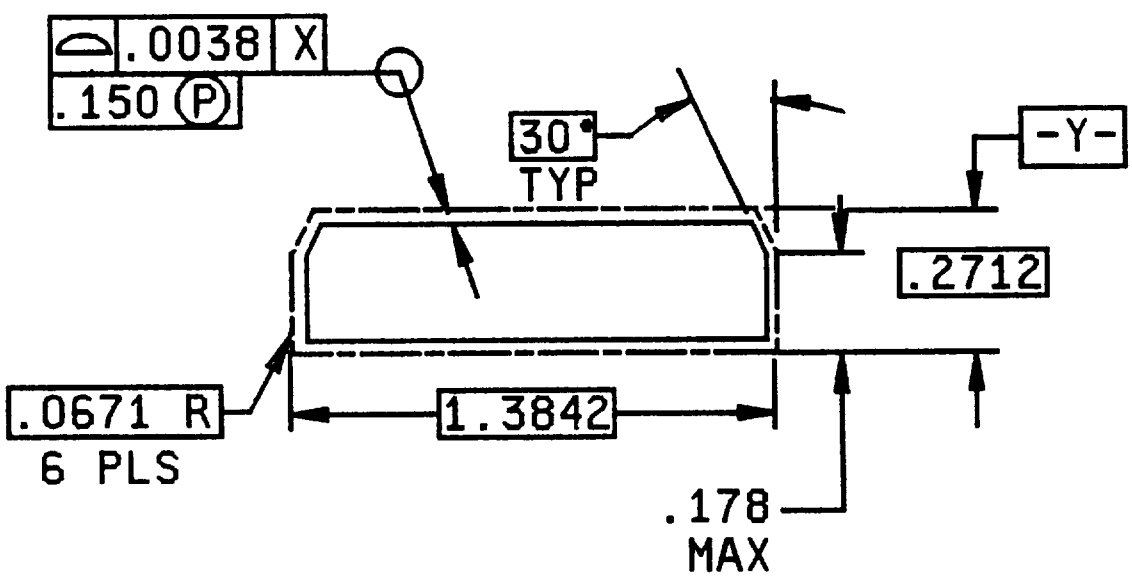


Inches	mm
.003	0.08
0038	0.097
.005	0.13
.007	0.18
.015	0.38
.025	0.64
050	1.27
.0669	1.699
.096	2.44
.140	3.56
.165	4.19
178	4.51
.185	4.70
.190	4.83
2278	5.786
.351	8.92
.355	9.02
.9838	24.989
1.215	30.86
1.775	45.09
2.000	50.80
2.27	57.66

FIGURE 1 Connector, plug, 050 spacing

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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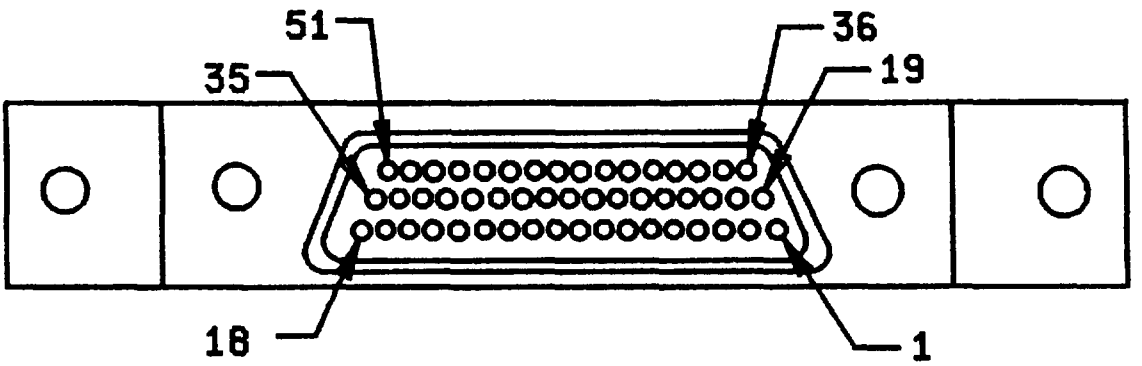


DETAIL OF INTERFACE

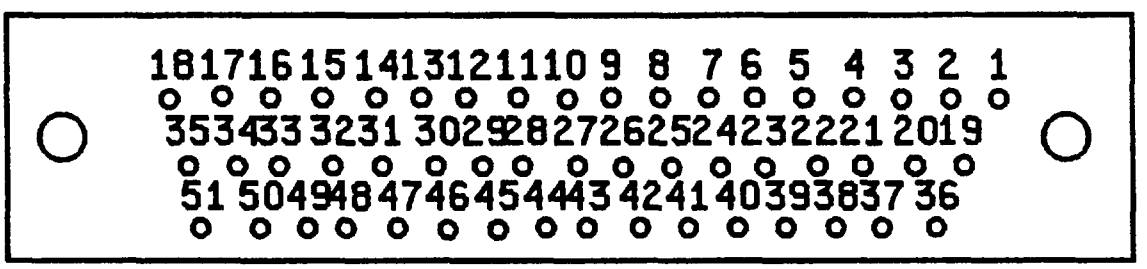
NOTES
1. Dimensions are in inches
2 Metric equivalents are given for information only
3 Unless otherwise specified, tolerances are ± 005 (0.13 mm).
4 Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513 Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519
5 Metal shell shall be of material in accordance with MIL-C-83513 for class M parts.
6 Jackpost (permanently attached), when specified· Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35
7 Separately molded plastic body (if used) shall conform to the requirements of type GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8 Wire termination pins shall conform to QQ-W-343, no 24 AWG copper
9 Termination lengths available .109 (2.77 mm), 140 (3 56 mm) or .172 (4 37 mm). The tolerance shall be ± 015 (0 381 mm) for all termination lengths

FIGURE 1 Connector, plug, 050 spacing - Continued

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51 CONTACT PLUG (PIN) CONNECTOR
VERTICAL MOUNT



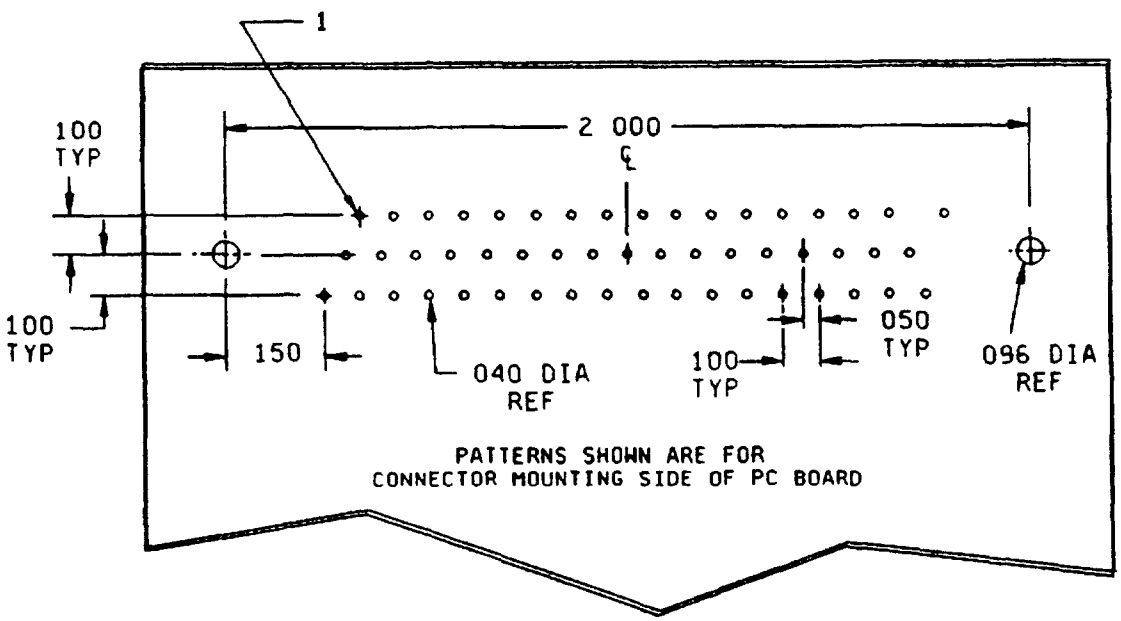
TERMINATION VIEW

NOTE

Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2. Insert arrangement

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Inches	mm
.040	1.02
.050	1 27
.096	2 44
.100	2.54
.150	3.81
2 000	63.50

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for information only

FIGURE 3 Layout arrangement

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REQUIREMENTS

Dimensions and configurations See figures 1 and 2

Current rating, maximum 3 amperes per contact

Materials

Termination organization area Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513

Shell The requirements for shell materials shall be in accordance with MIL-C-83513

Plastic body or plastic molding. Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.

Jackpost. Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35

Wire termination pins: Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper

Mating connector. Shall conform to MIL-C-83513/2 and MIL-C-83513/4

Plating of termination leads. Solder dipping of termination leads will be accomplished in Sn60 Pb40 or Sn63 Pb37 in accordance with QQ-S-571

Part Identification Number (PIN) Consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code for the shell finish and hardware column

M83513/23A-	G	01	C	P
Specification sheet number	Insert arrangements (see figure 2) G = 51	Termination length 01 = 109 02 = 140 03 = .172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities
Army - MI
Navy - AS, SH
Air Force - 17, 99
User activities
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent
DLA - ES

(Project 5935-3921-14)

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